

Sentence Production: Hybrid Layout and Magnitude-Independent Coding

Morgan, Devinsky, Doyle, Dugan, Friedman, Flinker (NYU Grossman School of Medicine / NYU Tandon), *Science Advances* 12(32), eaec0518, 5 August 2026. doi:10.1126/sciadv.aec0518

Takeaways

- In 10 epilepsy patients (1256 electrodes), two supposedly equivalent probes of "higher-order language" barely agree: sentence-vs-list flagged 60 electrodes, active-vs-passive 125, only 6 both.
- Measuring *information* instead of *activity* found sentence structure at 133 electrodes, event meaning at 92, word form at 126 — mostly different ones.
- Headline: structure and meaning content were unrelated to how hard an electrode fired (structure $P=0.987$, semantics $P=0.248$); word-form content tracked firing strength as expected ($P<0.001$).
- Clustering 741 electrodes gave 5 networks: three quiet, broadly distributed, high-information ones plus two loud ones locked to the picture and to speech.
- Layout is "hybrid," distributed *and* focal, so activity-thresholded localizers miss the sites carrying higher-order linguistic information.

Background

Almost all we know about language in the brain comes from *comprehension* studies in scanners: speaking creates motion and muscle artifacts that wreck fMRI and EEG. Production is poorly mapped, and the field is split between "syntax lives in specific hubs" and "syntax is spread across the language network." Lineage: (1) Sahin 2009 — word identity, grammar, and sound processed in that order within millimeters of Broca's area; (2) Pallier 2011 — response magnitude scaling with amount of structure, the logic this paper breaks; (3) Morgan 2025 — same lab and paradigm, decoded the planned word and its subject/object role.

What They Did

Ten patients (3 female, mean age 30, range 20–45) had electrode grids placed on the left hemisphere before epilepsy surgery — **ECoG** (electrocorticography): voltage read off the cortical surface, immune to speech motion artifacts, with millimeter and millisecond resolution. The measure is **high-gamma** power, the strength of 70–150 Hz fluctuations, tracking local firing and the fMRI signal.

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Sentence (60): question -> cartoon -> [PLAN ~1141 ms] -> speech
  active "Who sprayed whom?" -> "The chicken sprayed Dracula"
  passive "Who was sprayed by whom?" -> "Dracula was sprayed by the chicken"
List (60): arrow -> same cartoon -> "chicken Dracula" Naming (96): "chicken"
MAGNITUDE sentence > list -> 60 elec; active != passive -> 125; both: 6
INFORMATION (RSA) |dY| = b0 + b1*D_struct + b2*D_seman + b3*D_lex + b4*D_RT
                        133          92          126 electrodes
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Same picture, same meaning, same content words, different grammar. Analysis targets the **planning period**, picture onset to speech onset; passives took longer (median 1424 vs 1165 ms), so trials were time-warped to a common duration. **RSA** (representational similarity

analysis) is pairwise: for every two trials, compute how different the neural activity was, then regress that on how different the trials were in structure (active vs. passive), meaning (distance between GPT-2 layer-8 sentence embeddings), and word form (same vs. different first word), with reaction-time difference as a covariate.

What They Found

1. Hybrid spatial organization. Neither side of the localization debate is right: information sits across broad swaths of left peri-Sylvian cortex *and* concentrates in specific regions.

Information	Term	Focal regions	Tracks activity?
Sentence structure	D_struct	IFG, MFG	No ($P = 0.987$)
Event meaning	D_seman	MTG, IPL	No ($P = 0.248$)
Word form / sounds	D_lex	STG, SMC	Yes ($P < 0.001$)

IFG/MFG = inferior/middle frontal gyrus; STG/MTG = superior/middle temporal gyrus; IPL = inferior parietal lobule; SMC = sensorimotor cortex. Most electrodes carried only one of the three, arguing against a single interwoven system. Non-negative matrix factorization over 741 electrodes recovered the split with no anatomy given: three clusters are broadly distributed and near baseline in activity, each specialized for one information type, while the two loud clusters — frontal at picture onset, sensorimotor at speech onset — carry no structure or meaning information ($P > 0.05$).

2. Magnitude-independent coding. The field assumes — the authors call it *elevation-processing equivalence* — that a region working harder shows more activity. That fails for higher-order language. Mechanically: an electrode can sit at its resting mean all task long, yet its small trial-to-trial deviations around that mean are reliably larger between an active and a passive sentence than between two actives. The signal lives in the *pattern of differences across trials*, not the *loudness*. Deep-learning analogue: a hidden unit whose mean activation is identical across classes but whose activation *direction* is class-informative — a linear probe reads it out, a mean-activation heatmap shows nothing. Word form is the control that behaves the old way, making this a dissociation, not a null result.

Why It Matters

Much of language neuroscience finds regions by thresholding on activity, then studies only those voxels or channels. If higher-order information is magnitude-independent, that pipeline discards the relevant sites before analysis begins. For speech brain-computer interfaces this is a hypothesis, not a result — the paper neither cites nor tests BCI work. Current neuroprostheses decode from loud sensorimotor sites, exactly where word-form information lives here, so that choice looks well-founded; but structure and meaning at quiet distributed sites would be invisible to activity-driven channel selection.

On the AI parallel, stay honest. GPT-2 embeddings appear only as an off-the-shelf meaning space, built from active-syntax sentences so the regressor stayed orthogonal to the structure term. Nothing here says the brain is a transformer. The real overlap is methodological — both fields moved from "which unit lights up" to "what can be linearly read out." The divergence is as real: a transformer carries syntax and semantics in one residual stream at the same layers; this study finds them at non-overlapping cortical sites.

Key Papers

1. Morgan 2026, *Sci. Adv.* — this paper. doi · preprint · data

2. Sahin 2009, *Science* — staged lexical/grammatical/phonological processing in Broca's area. doi · Morgan 2025, *Comms Psych* — decoding words during sentence production. doi
3. Pallier 2011, *PNAS* — activity scaling with constituent structure. doi · Matchin & Hickok 2020 — syntax-hub position. doi · Kriegeskorte 2008 — RSA method. doi · Grill-Spector 2006 — repetition suppression. doi